Cumulant proxies versus RBM+TDVP infidelity in 2D quenches

Execution report of `PLAN_quench_family.md` (Part I)
and `PLAN_rydberg_quench.md` (Part II)

27 October 2026

Abstract

We test whether the cumulant proxy $\mu(t)$, the position of a Fermi–Dirac fit to the ranked Walsh–Hadamard spectrum of $\log|\psi(s,t)|$, predicts when a restricted Boltzmann machine (RBM) evolved with full-sum TDVP stops following an exact quench in two dimensions. *Part I* studies two quench families from the transverse-field Ising (TFI) paramagnet on a 4×3 cylinder: a field quench and a quench that switches on an XY coupling. The proxy onset time orders the RBM failure time with Spearman correlations 0.91 and 0.99. At fixed half-cut entropy S, μ keeps a partial correlation of 0.81 with the infidelity, whereas S keeps only 0.32 at fixed μ. The proxy saturates for hard quenches, where the phase proxy μ_{ph} and a fit-free high-order weight do slightly better. *Part II* screens 37 Rydberg, dipolar, trapped-ion and benchmark quench families (976 points, 159 RBM runs) for an *RBM advantage*: states that are hard for matrix product states (MPS) but easy for the RBM. Two families carry the result. In the weakly driven 2D Rydberg Ising quench the RBM keeps the infidelity below 10^{-3} for 1.6–12 times the time the entanglement needs to become MPS-hard, also at $N = 16$ (near-classical RBM advantage). In the 2D TFI benchmark quench, and in every other genuinely non-classical family, proxy onset, RBM failure and entanglement saturation coincide: we found no quantum RBM advantage.

1 Overview

- *Question.* Does the cumulant proxy $\mu(t)$ (Sec. 2), computed from the exact state alone, predict when an RBM evolved with TDVP stops following a 2D quench, and does it do better than the entanglement entropy S?
- *Method.* Exact evolution on 12–16 sites, the proxies at every time step, and full-sum RBM+TDVP runs whose infidelity with the exact state is the ground truth.
- *Part I* (Secs. 3–6): a controlled test with two quench families (field quench, XY-coupling quench) on a 4×3 cylinder (1D calibration and a 2D parameter scan in Apps. A, B). Answer: yes. μ orders the RBM failure times and carries information beyond S, but it saturates for hard quenches.
- *Part II* (Secs. 7–11): a screen of experimentally relevant quenches for states that are MPS-hard but RBM-easy, discussed through two families: the 2D Rydberg Ising quench (R1) and the 2D TFI benchmark (G4); all other families are in Apps. C–E. Answer: only near-classical quenches give an RBM advantage; there is no quantum RBM advantage.

2 Definitions

All quantities refer to N spins in the σ^z basis, $s \in \{\pm 1\}^N$, $\chi_A(s) = \prod_{i \in A} s_i$.

Cumulant (cluster) proxies.

- $c_A = 2^{-N} \sum_s \chi_A(s) \log|\psi(s)|$: cluster coefficients of $\log|\psi|$ (Walsh–Hadamard transform); $|A|$ is the order of the cluster. Coefficients with $\log|c_A| < -28$ are noise ("symmetry zeros").

- μ: Fermi–Dirac fit $\mathrm{rank}(x) = A_0/(e^{(x-\mu)/T}+1)$ of the rank of $|c_A|$ against $x = \log|c_A|$, with $A_0 = N_{\mathrm{real}}$ (number of non-noise coefficients). μ is the log-size at which the spectrum stops being dense; small μ means few relevant clusters. If no coefficient of order > 2 survives the noise cut, $\log|\psi|$ is exactly two-body and we set $\mu = -\infty$ (shown at the floor -30).
- μ_{ph}: the same fit applied to the phase $\varphi(s,t) = \arg\psi(s,t)$, unwrapped continuously in t, with its mean removed.
- $w_{\mathrm{hi}} = \sum_{|A|>4} c_A^2 / \sum_{|A|\geq 1} c_A^2$: fit-free weight of the high-order clusters.
- *Auxiliary*, reported only where noted: μ_2 (the same fit with $A_0 = 2N_{\mathrm{real}}$), the fit width T, the near-node indicator $\mathrm{minlog} = \min_s \log(2^N|\psi(s)|^2)$, and $\tilde{\mu}_{\mathrm{ph}}$ (μ_{ph} of the interaction-picture phase $\varphi + \int_0^t E_d(s,t')dt'$, $E_d(s) = \langle s|H|s\rangle$).

Entanglement and MPS cost.

- $S = -\mathrm{tr}\rho_L \log\rho_L$ (nats), with L the first $N/2$ sites of the site ordering (half cut).
- $S_{\mathrm{Page}} = \frac{N}{2}\log 2 - \frac{1}{2}$: the mean half-cut entropy of a random state; $S_{\mathrm{rel}} = S/S_{\mathrm{Page}}$.
- χ_ε: the smallest number of Schmidt values whose discarded weight is $\leq \varepsilon = 10^{-3}$, at the cut with the largest S. Its cap at that cut is $2^{\min(\ell, N-\ell)}$; we report the fraction $(\chi_\varepsilon - 1)/(\mathrm{cap} - 1)$, which is 0 for a product state and 1 at the cap.
- n_∂: number of nearest-neighbour bonds cut by the half cut.

Non-classicality.

- $\eta = \|H - \mathrm{diag}_{\sigma^z} H\|_F / \|H - \mathrm{tr}H/D\|_F$ ($D = 2^N$): 0 for an Ising (diagonal) Hamiltonian, 1 for a purely off-diagonal one. For Floquet protocols, H is the period-averaged Hamiltonian.

Times and windows. "Never" means that a threshold is not crossed within the run.

- t^*_μ, t^*_{ph}: first time with $\mu > -10$, $\mu_{\mathrm{ph}} > -10$ (absolute threshold, Part II); $t^*_{\mathrm{prox}} = \min(t^*_\mu, t^*_{\mathrm{ph}})$.
- $t^*_{\mu,\mathrm{rel}}$: first time with $\mu > \mu_0 + 5$, $\mu_0 = \mu(0)$ (relative threshold, Part I).
- t^*_w: first t with $w_{\mathrm{hi}} > 1\%$; t_{dev}: first t with $\mu > \mu_0 + 1.5$ (end of the plateau of μ).
- t_S: first t with $S_{\mathrm{rel}} \geq 0.5$ ("MPS-hard" at this size).
- $I = 1 - |\langle\psi_{\mathrm{ED}}|\psi_{\mathrm{RBM}}\rangle|^2/(\|\psi_{\mathrm{ED}}\|^2\|\psi_{\mathrm{RBM}}\|^2)$: RBM infidelity; t^*_{inf}: first t with $I > 10^{-3}$.
- *Proxy window* t^*_{prox}/t_S and *RBM window* t^*_{inf}/t_S: how long after the state becomes MPS-hard the proxies stay quiet, respectively the RBM stays accurate, in units of t_S.

Ansätze. $\log\psi(s) = \sum_i a_i s_i + \sum_{j=1}^{\alpha N} \log\cosh\left(b_j + \sum_i W_{ji} s_i\right) + \sum_{i<j} K_{ij} s_i s_j$, all parameters complex. "RBM α": $K = 0$ with αN hidden units; "Jastrow": no hidden units; "RBM×Jastrow": $\alpha = 1$ and K.

Part I
TFI/XXZ quench family

3 Setup

Lattice and basis. $N = 12$ spins on a 4×3 lattice, periodic along x ($L_x = 4$) and open along y ($L_y = 3$), so that the nearest-neighbour graph stays bipartite. Sites are labelled $i = xL_y + y$, and we use the σ^z basis in NetKet ordering for both ED and the RBM.

Quench families. The initial state is the ground state of the 2D TFI paramagnet,

$$|\psi_0\rangle = \mathrm{gs}\Big(-\sum_{\langle ij\rangle} Z_i Z_j - h_0 \sum_i X_i\Big), \qquad h_0 = 4, \tag{1}$$

and it is evolved with

$$H_f = -J\sum_{\langle ij\rangle} Z_i Z_j + J_{xx}\sum_{\langle ij\rangle}\big(X_i X_j + Y_i Y_j\big) - h_f \sum_i X_i, \qquad J = 1. \tag{2}$$

- **Field quench** (baseline): $J_{xx} = 0$, control parameter $h_f \in [0.2, 4]$ (20 values). This is the 2D analogue of the existing 1D study (calibration in App. A). The bulk 2D critical point is $h_c \approx 3.04$. On this small cluster it is only a crossover.
- **XY-coupling quench** (primary): $h_f = 1$, control parameter $J_{xx} \in [-1.5, 1.5]$ (31 values). It interpolates between the TFI ($J_{xx} = 0$), the SU(2) ferromagnet ($J_{xx} = -1$), and the easy-axis / XY-antiferromagnetic regime ($J_{xx} > 0$).

The two families share one point, $(J_{xx}, h_f) = (0, 1)$. It gives the same ED and RBM data in both families, with $I = 0.407$ at $t = 1.5$, which is a consistency check of the pipeline.

Proxies. For each t on the grid $t \in [0, 1.5]$, $dt = 0.01$, we compute the exact $\psi(t)$ by full diagonalisation and the c_A by a fast Walsh–Hadamard transform, drop $c_\emptyset$ and the symmetry zeros (spin-flip $\mathbb{Z}_2$ removes every odd cluster), and fit μ, μ_{ph} and the auxiliary proxies (Sec. 2). Each fit is warm-started from the previous time step; fits that run away are rejected. All 147 ED parameter points ($31 + 20 + 96$ points of the plane scan in App. B) were fitted without a single rejected fit, with $R^2 \geq 0.93$ throughout.

RBM+TDVP. The ansatz is a complex RBM with $\alpha = 1$ (168 parameters), plus $\alpha = 2$ at five points, evolved with full-sum TDVP and identical solver settings for every run (App. F). The initial RBM comes from 1000 SR steps of infidelity minimisation against ψ_0, which gives an initial infidelity of 4.8×10^{-6} ($\alpha = 1$) and 3.8×10^{-6} ($\alpha = 2$). We ran 46 TDVP runs: the XY-coupling quench on all 31 points, the field quench on 10 points, and $\alpha = 2$ at 5 points. Each $\alpha = 1$ run took 5–10 min on one core; none crashed.

Onset times. Part I uses the relative onset time $t^*_{\mu,\mathrm{rel}}$ ($\mu > \mu_0 + 5$, with $\mu_0 = -14.63$) as the primary proxy onset, together with t^*_w, t_{dev} and t^*_{inf} (Sec. 2). "Never" means not crossed within $t \leq 1.5$.

4 Exact dynamics of the proxies

The field quench shows a clean "plateau then jump"; the XY-coupling quench has a 10× range of onset times but almost no plateau.

Field quench (Figs. 1, 2b).

- μ stays flat (or even dips slightly) up to $t_{\mathrm{dev}} \approx 0.18$–$0.35$ for all h_f.
- For quenches deep into the ferromagnetic side ($h_f \lesssim 2.4$), μ then rises to ≈ -5.
- $t^*_{\mu,\mathrm{rel}}$ is shortest (0.24) around $h_f \approx 1$–1.4 and grows towards both $h_f \to 0$ (0.57) and $h_f \to h_c$ (0.53 at $h_f = 3$).
- For $h_f \geq 3.2$, μ never crosses threshold.

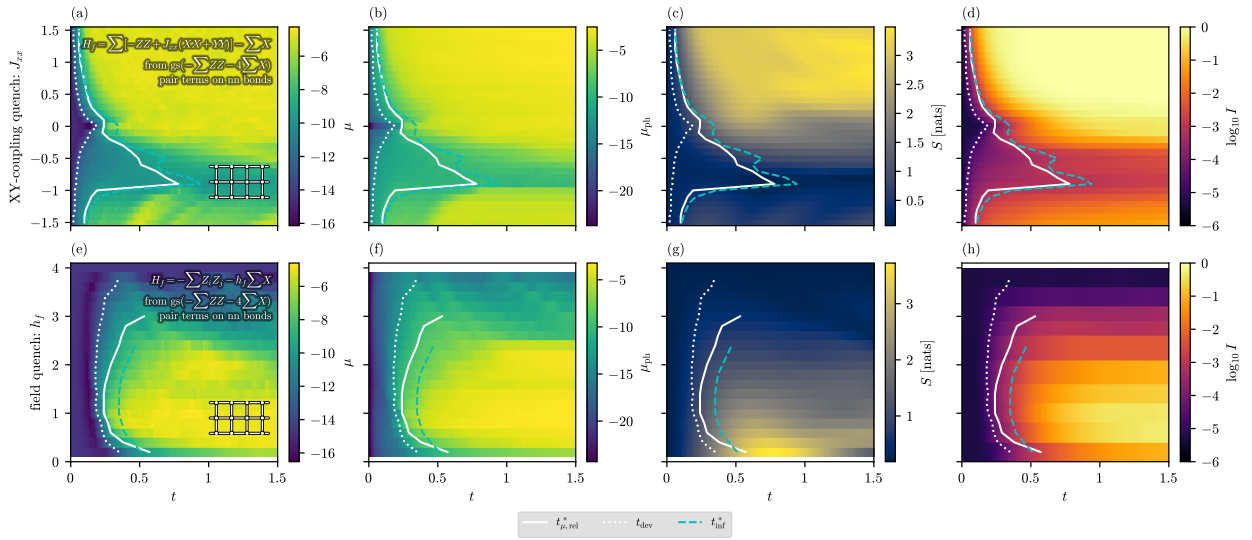


Figure 1: Proxies of the exact state next to the RBM infidelity, over time t and control parameter. Top row: XY-coupling quench ($h_f = 1$, control J_{xx}). Bottom row: field quench ($J_{xx} = 0$, control h_f). Columns: cumulant proxy μ of $\log|\psi|$, phase proxy μ_{ph} (white at $h_f = 4 = h_0$, where ψ_0 is an eigenstate and $\mu_{\mathrm{ph}} = -\infty$), half-cut entanglement entropy S, and $\log_{10}$ of the infidelity I of the $\alpha = 1$ RBM (field quench: only the 10 values of h_f where TDVP was run). Lines: relative onset time $t^*_{\mu,\mathrm{rel}}$ ($\mu > \mu_0 + 5$, white solid), end of the plateau t_{dev} ($\mu > \mu_0 + 1.5$, white dotted), RBM failure time t^*_{inf} ($I > 10^{-3}$, cyan dashed). White insets (here and in later heatmaps): final Hamiltonian, initial state and the 4×3 cylinder (stubs: bonds across the periodic boundary).

XY-coupling quench (Figs. 1, 2a). $t^*_{\mu,\mathrm{rel}}$ runs from 0.06 at $J_{xx} = 1.5$ to 0.78 at $J_{xx} = -0.9$. The plateau, however, is short:

- $t_{\mathrm{dev}} \leq 0.07$ whenever $|J_{xx}| \geq 0.4$, because a sudden XY coupling acts immediately on the σ^z amplitudes;
- near the SU(2) ferromagnet ($J_{xx} \in [-0.9, -0.6]$), μ jumps quickly to ≈ -10 and then *plateaus* there, just at the threshold;
- for $J_{xx} > 0$, μ saturates at ≈ -5 within $t \lesssim 0.25$, while S keeps growing to 3.5 (out of a maximum of 4.2).

$t^*_{\mu,\mathrm{rel}}$ is discontinuous between $J_{xx} = -1.0$ (0.19) and -0.9 (0.78). The reason is that the plateau value of μ sits at the threshold, so a small change in the plateau moves $t^*_{\mu,\mathrm{rel}}$ by a lot.

A scan of the full (J_{xx}, h_f) plane (App. B) shows that a larger h_f widens the range of onset times but does not lengthen the plateau.

Robustness (Fig. 3). The time step is sufficient; the relative threshold is not robust to the initial state.

- *Phase unwrapping.* Halving dt changes μ and μ_{ph} by less than 3×10^{-6} at $J_{xx} = -0.9$ and 0. At $J_{xx} = 0.9$ the changes are 0.008 (μ) and 0.16 (μ_{ph}), so $dt = 0.01$ is sufficient.
- *Initial field.* Raising h_0 to 6 lowers μ_0 from -14.6 to -17.9. The late-time values are nearly unchanged for hard quenches. The relative threshold $\mu_0 + 5$, however, is then crossed earlier, and in two cases at $t \approx 0.2$ instead of late or never:
  - $J_{xx} = -0.9$: $t^*_{\mu,\mathrm{rel}} = 0.07$ instead of 0.78;
  - $h_f = 3.6$: $t^*_{\mu,\mathrm{rel}} = 0.21$ instead of never.

  An absolute threshold ($\mu > -10$, used in Part II) is more transferable.

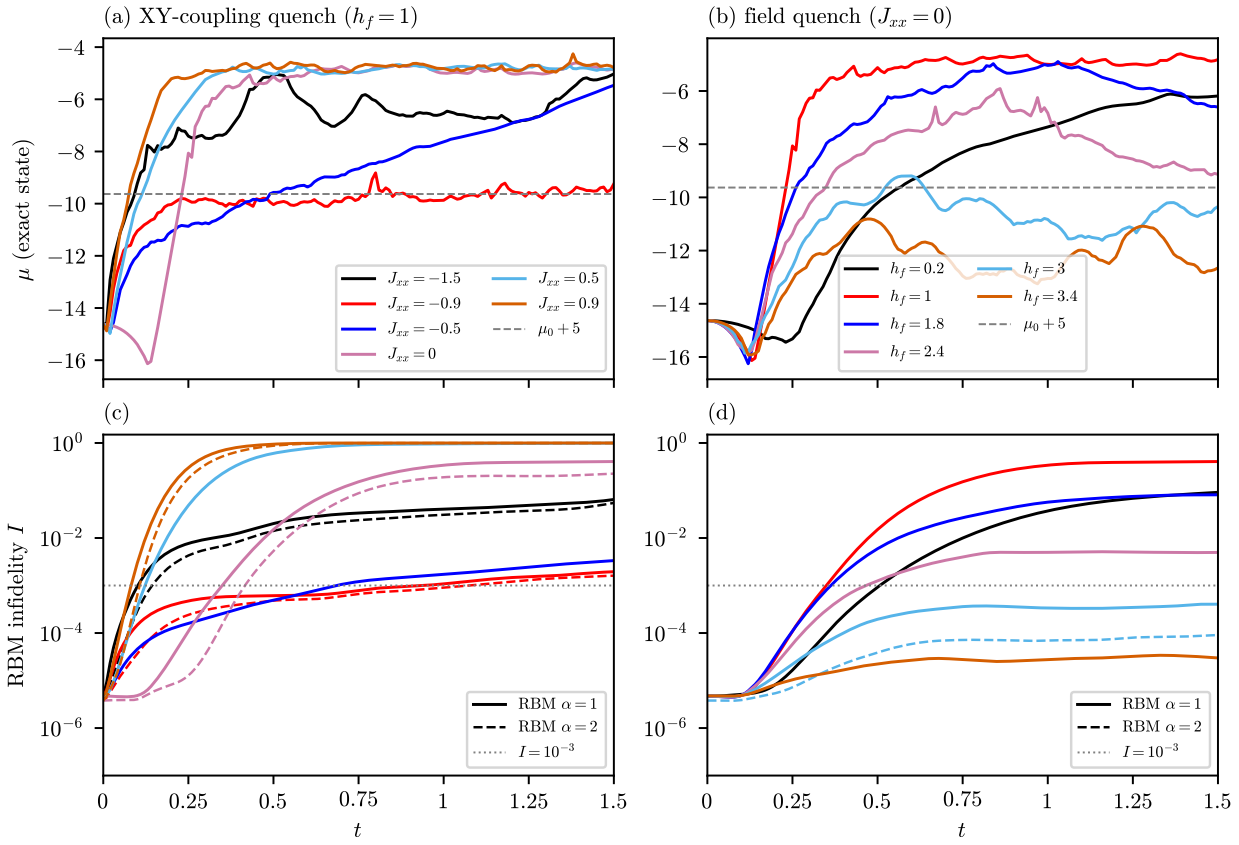


Figure 2: Cumulant proxy of the exact state (top) and RBM infidelity (bottom) along time, at representative control parameters; equal colours mark the same quench. Left: XY-coupling quench ($h_f = 1$, control J_{xx}). Right: field quench ($J_{xx} = 0$, control h_f). (a,b) $\mu(t)$; grey dashed: threshold $\mu_0 + 5$ that defines the relative onset time $t^*_{\mu,\mathrm{rel}}$. (c,d) Infidelity I of the RBM with $\alpha = 1$ (solid) and $\alpha = 2$ hidden units per spin (dashed, run at $J_{xx} \in \{-1.5, -0.9, 0, 0.9\}$ and $h_f = 3$); grey dotted: $I = 10^{-3}$, which defines the failure time t^*_{inf}. Both quenches start from the ground state of the 2D TFI paramagnet ($h_0 = 4$) on the 4×3 cylinder.

5 RBM infidelity versus the proxy

μ orders the RBM failure times in both quench families and predicts the infidelity beyond S; it loses resolution once it saturates at ≈ -5.

Onset times. The proxy and the RBM infidelity agree qualitatively over the whole (t, p) plane (Fig. 1, right column). Fig. 4 compares the onset times quantitatively.

- *XY-coupling quench.* $t^*_{\mu,\mathrm{rel}}$ and t^*_{inf} follow each other almost point by point (Spearman 0.99 across J_{xx}). This includes the sharp maximum at $J_{xx} = -0.9$ ($t^*_{\mu,\mathrm{rel}} = 0.78$, $t^*_{\mathrm{inf}} = 0.94$). Typically $t^*_{\mathrm{inf}} \gtrsim t^*_{\mu,\mathrm{rel}}$, i.e. μ gives a slightly early warning.
- *Field quench.* The Spearman correlation is 0.91.
  - The two "never" points of μ ($h_f = 3.4$ and 3.8) end at $I = 3.0 \times 10^{-5}$ and 5.9×10^{-6}.
  - $h_f = 2.8$ and 3.0 are marginal: μ just crosses the threshold at $t^*_{\mu,\mathrm{rel}} = 0.40$ and 0.53, while I saturates just below 10^{-3} (7.8×10^{-4} and 4.0×10^{-4}).
- *Fit-free proxy.* t^*_w ranks even better in the field quench (Spearman 0.98) and worse in the XY-coupling quench (0.95). It flags the SU(2) region of the latter as never hard, while the infidelity saturates at 2–3 $\times 10^{-3}$ there.

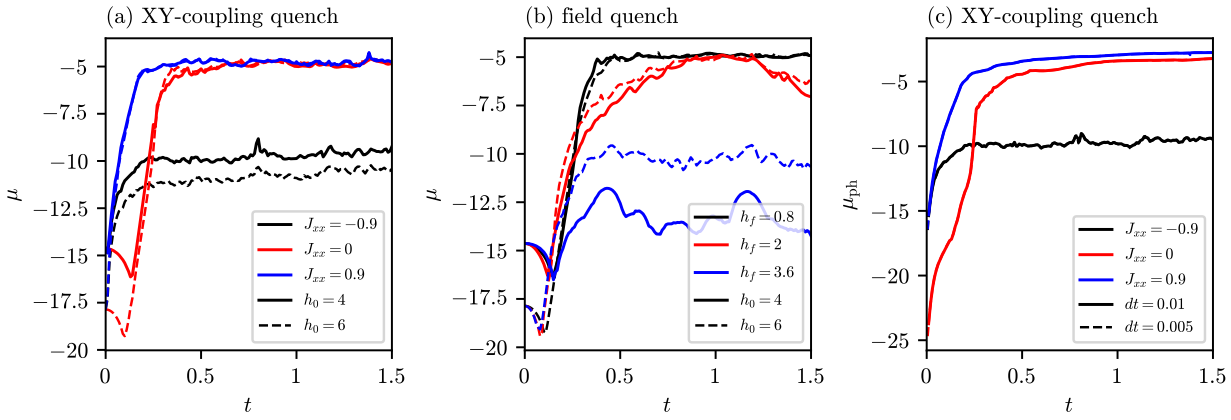


Figure 3: Robustness. (a,b) $\mu(t)$ for initial fields $h_0 = 4$ (solid) and $h_0 = 6$ (dashed). (c) $\mu_{\rm ph}$ with $dt = 0.01$ (solid) and $dt = 0.005$ (dashed).

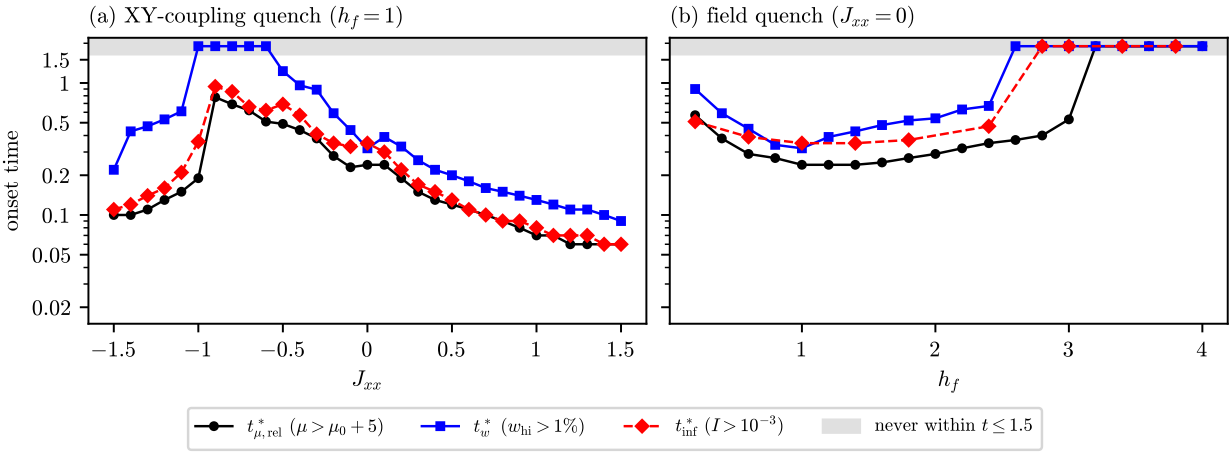


Figure 4: Onset times against the control parameter. Points in the grey band never cross their threshold for $t \le 1.5$.

Pointwise relation and entanglement. The scatter plots (Fig. 5) show that $\log I$ is close to a monotone function of μ for $\mu \lesssim -8$. Above that, μ saturates while the infidelity spreads over two decades. Within that spread, a higher S means a larger infidelity (panel a). This is where μ loses resolution. The S panels (b,d) show the opposite structure: at fixed S, the infidelity spans up to 3.5 decades (e.g. 5×10^{-6} to 10^{-2} at $S \approx 0.3$ in the XY-coupling quench), ordered by μ. The two proxies are strongly correlated with each other (Spearman$(\mu, S) = 0.82$ for the field quench and 0.90 for the XY-coupling quench), so we quantify their separate content with partial correlations (Table 1).

Correlation analysis. Table 1 gives the key test of the plan.

- After removing S, μ keeps a large partial correlation with the infidelity: 0.86 (field), 0.65 (XY coupling), 0.81 (pooled). This survives additionally removing t (0.78 pooled).
- The converse is much weaker. Once μ is known, S adds only 0.33 (field) and 0.49 (XY coupling). So μ carries most of the information in S about RBM hardness, and more besides.
- $\mu_{\rm ph}$ and $w_{\rm hi}$ do best (pooled $\rho|S \approx 0.92$), and they retain 0.50–0.82 even after controlling for μ (higher in the XY-coupling quench). This mainly reflects their resolution in the regime where μ has saturated.
- The auxiliary proxies μ_2, T and minlog add nothing beyond μ ($|\rho(\cdot|\mu)| \le 0.28$). The near-node

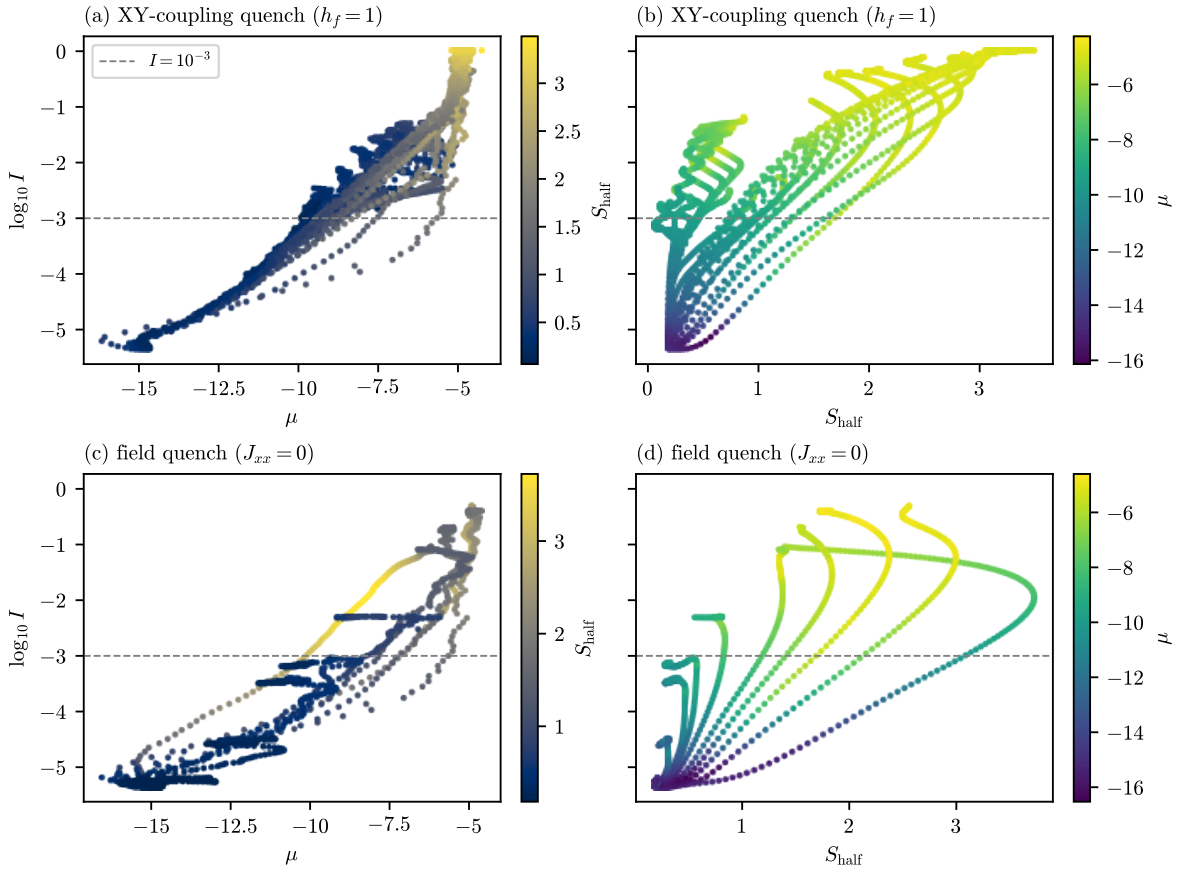


Figure 5: Every (t, p) point with $t > 0$. (a,c) $\log_{10} I$ against μ, coloured by S. (b,d) the same against S, coloured by μ.

indicator, which drives the jump in 1D, also carries almost no information independent of S in 2D ($\rho|S = -0.37$ for the field quench, -0.06 for the XY-coupling quench).

Larger RBM. Doubling the hidden units (Fig. 2c,d, dashed lines) does not change the picture.

- *Easy and intermediate quenches* improve by a factor of 1.2–4.4:
  - $h_f = 3$: $4.0 \times 10^{-4} \to 9.1 \times 10^{-5}$;
  - $J_{xx} = -0.9$: $2.0 \times 10^{-3} \to 1.6 \times 10^{-3}$;
  - $J_{xx} = 0$: $0.41 \to 0.23$.
- *Hard quenches* are essentially unaffected: at $J_{xx} = 0.9$, I exceeds 0.9 at $t = 0.46$ for $\alpha = 1$ and at $t = 0.52$ for $\alpha = 2$.

In the high-μ regime, the failure is therefore not cured by modestly increasing the expressivity.

6 Verdict of Part I

All three success criteria of the plan are met (Table 2).

Where μ works.

- It predicts the RBM failure time t^*_{inf} across a 10× range of onset times, and it usually fires slightly before the infidelity crosses 10^{-3}.

Table 1: Spearman correlation ρ of each proxy with $\log_{10} I$ over all (t, p) with $t > 0$ ($n = 1500$ for the field quench, 4650 for the XY-coupling quench), and partial Spearman correlations after linearly removing the ranks of S, of (S, t), or of μ. Consecutive time steps are strongly autocorrelated, so the effective sample size is far below n and no p-values are given.

	proxy	ρ	$\rho\|S$	$\rho\|S,t$	$\rho\|\mu$		proxy	ρ	$\rho\|S$	$\rho\|S,t$	$\rho\|\mu$
field	μ	0.96	0.86	0.86	–	XY	μ	0.94	0.65	0.59	–
	μ_{ph}	0.97	0.92	0.89	0.62		μ_{ph}	0.98	0.88	0.83	0.82
	w_{hi}	0.97	0.89	0.87	0.50		w_{hi}	0.98	0.86	0.83	0.80
	S	0.84	–	–	0.33		S	0.91	–	–	0.49
Pooled: $\rho(\mu) = 0.96$, $\rho(\mu\|S) = 0.81$, $\rho(\mu\|S,t) = 0.78$; $\rho(S\|\mu) = 0.32$; $\rho(\mu_{\mathrm{ph}}\|S) = 0.92$; $\rho(w_{\mathrm{hi}}\|S) = 0.91$.											

Table 2: Success criteria of `PLAN_quench_family.md`.

criterion	result	met
$t^*_{\mu,\mathrm{rel}}(p)$ and $t^*_{\mathrm{inf}}(p)$ agree in ordering, Spearman ≥ 0.8	field: 0.91; XY coupling: 0.99	yes
"Never hard" points in μ have $I < 10^{-3}$ at $t = 1.5$	field, $h_f = 3.4$: 3.0×10^{-5}; $h_f = 3.8$: 5.9×10^{-6} (XY coupling has no never points)	yes
μ predicts beyond S	partial $\rho(\mu\|S) = 0.81$ pooled (field 0.86, XY coupling 0.65) vs $\rho(S\|\mu) = 0.32$	yes

- It correctly identifies the slow region around the SU(2) ferromagnet. There, μ and I both saturate at a low plateau ($\mu \approx -9.5$, $I \approx 2 \times 10^{-3}$).
- It correctly identifies the paramagnetic side of the field quench as easy.
- It carries information on RBM hardness that the entanglement entropy does not.

Where μ fails or is fragile.

1. *Saturation.* μ caps at ≈ -5 once the spectrum is dense, so it cannot rank hard quenches against each other. Over the saturated region, I ranges from 0.06 to 1. S, μ_{ph} and w_{hi} keep resolving it.
2. *Strong ferromagnetic XY ($J_{xx} = -1.5$).* μ reaches -5 (as "hard" as the TFI quench at $J_{xx} = 0$), but the infidelity only reaches 0.065, against 0.41 at $J_{xx} = 0$. The low entanglement ($S = 0.87$) is the better guide here. This is the decoupling point flagged in the plan, and at it μ *overestimates* the difficulty.
3. *No plateau in the XY-coupling quench.* With $|J_{xx}| \geq 0.4$, μ starts moving after $t_{\mathrm{dev}} = 0.02$–0.07, while I first exceeds 10^{-4} only at $t = 0.04$–0.17, i.e. 2–3× later. Only the field quench has a real "flat then jump" plateau.
4. *Threshold dependence.* $t^*_{\mu,\mathrm{rel}}$ is defined relative to μ_0, so it is sensitive to the initial state (Fig. 3). It is also discontinuous where μ plateaus close to threshold ($J_{xx} = -1.0$ versus -0.9).

Recommendation.

- Use the XY-coupling quench ($h_f = 1$, control J_{xx}) as the primary family: it has the widest and most faithfully tracked range of t^*_{inf}.
- Keep the field quench as the reference for the clean plateau-then-jump and "never hard" behaviour.
- For a quantitative hardness predictor beyond the onset time, report μ_{ph} or the fit-free w_{hi} alongside μ. They do not saturate as early and have the highest partial correlations.

- Replace the relative threshold by an absolute one (e.g. $\mu > -10$) before comparing different initial states.

Limitations.

- *System size.* With 12 sites and a 3-site open direction, DQPT-like features are smeared. $h_c(4 \times 3)$ is not the bulk 3.04, and every t^* is a crossover time.
- *Error sources.* The TDVP infidelity mixes expressivity with integration and QGT-inversion error. The solver settings were kept fixed across p, but they were not varied.
- *Correlation statistics.* The correlations (here and in Part II) pool strongly autocorrelated time series, so the numbers are descriptive and not significance tests.
- *Threshold choices.* The thresholds $\mu_0 + 5$, 1% and 10^{-3} follow the plan and were not tuned.

Part II
Rydberg and related quenches

7 Targets and approach

Part II (`PLAN_rydberg_quench.md`) asks whether there are quenches in which the entanglement grows fast, so that the state becomes expensive for MPS, while the RBM+TDVP infidelity stays small. We call this an *RBM advantage* and distinguish two targets:

- *Near-classical RBM advantage*: H_f is close to diagonal in σ^z ($\eta \lesssim 0.1$). The amplitudes barely move and the phase is generated by two-body Ising terms, which an RBM with enough hidden units, or a Jastrow factor, can encode exactly, while the half-cut entanglement of such phase states is volume-law.
- *Quantum RBM advantage* ("gold" in the plan): the same with a genuinely non-classical H_f ($\eta \geq 0.3$), and with μ and μ_{ph} staying below -10 until t_S.

Approach.

- *Screen.* 37 quench families (976 parameter points) realised in Rydberg, dipolar and trapped-ion experiments, NQS benchmarks and theory, evolved exactly on 12–16 sites, with the proxies and the entanglement at every time step (Apps. C, D).
- *RBM runs.* Six Tier-1 families followed with 159 full-sum RBM/Jastrow+TDVP runs (App. E).
- *Main text.* Two families carry the conclusions: R1, the best near-classical case, and G4, a standard NQS benchmark and the closest approach to a quantum RBM advantage.

The two highlighted families. Both run on the 4×3 cylinder of Part I (R1 also on the 4×4 torus). With $n_i = (1 - Z_i)/2$ and $V_{ij} = V_{nn}(a/r_{ij})^6$, the Rydberg Hamiltonian is

$$H_{\mathrm{Ryd}} = \frac{\Omega}{2} \sum_i X_i - \Delta \sum_i n_i + \sum_{i<j} V_{ij} n_i n_j, \tag{3}$$

with Rabi frequency Ω, detuning Δ, and time in units of $1/V_{nn}$.

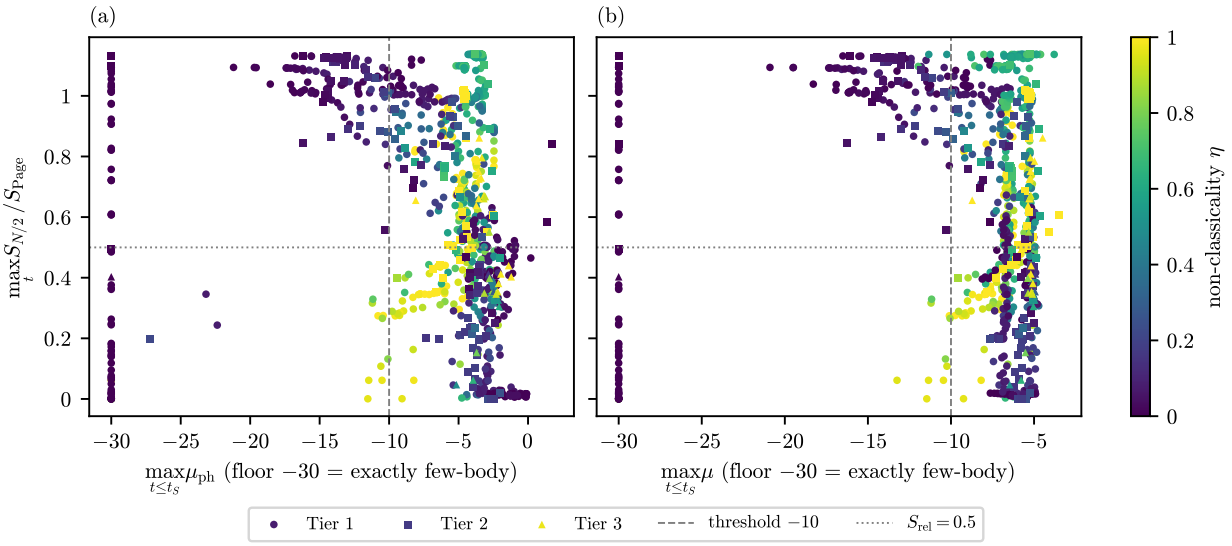


Figure 6: Every screened point (976 points, 37 families, 81 cluster/initial-state combinations), placed by its largest half-cut entropy relative to Page (vertical) and by the largest value of (a) μ_{ph}, (b) μ reached up to t_S (or over the whole run if S_{rel} never reaches 0.5). Colour: non-classicality η. The yellow upper-left region (MPS-hard, proxies below −10) is where a quantum RBM advantage would sit; it is empty. The points at the floor −30 are exactly classical.

- **R1, 2D Rydberg Ising quench** [1, 2]: H_{Ryd} with $\Delta = 0$ from $|{+}x\rangle$, control parameter Ω/V_{nn} from 0 (classical) to 1 (13 values), $t \le 3/\Omega$. Small Ω is near-classical ($\eta = 0.005$ at $\Omega/V_{nn} = 0.01$, 0.45 at 1).
- **G4, 2D TFI benchmark** [3]: Schmitt and Heyl simulated 2D TFI quenches with convolutional NQS at $h_f = h_c/10$, h_c and $2h_c$ ($h_c = 3.04438$). We use $H = -\sum_{i<j}(a/r_{ij})^6 Z_i Z_j - h_f \sum_i X_i$ from the paramagnetic ground state of Part I ($h = 4$), with 13 values of h_f including these three, $t \le 10$. It is non-classical for $h_f \gtrsim 0.6$ ($\eta = 0.42$–0.98).

RBM runs. An $\alpha = 1$ RBM over the full grid; at three points per family $\alpha = 2$, $\alpha = 4$ and a pure Jastrow factor; RBM×Jastrow at R1 $\Omega/V_{nn} = 0.05$; and an $\alpha = 1$ RBM on the 4×4 torus ($N = 16$) for R1. TDVP sums over all 2^N configurations, so there is no Monte Carlo noise. Product starts are exact through the visible biases (initial $I \approx 10^{-9}$); the ground-state start of G4 is fitted by SR (initial $I \approx 3 \times 10^{-6}$). Runs stop once $I \ge 0.5$. Solver settings: App. F.

8 Screen: near-classical windows, no quantum RBM advantage

The screen finds many near-classical windows and no candidate for a quantum RBM advantage (Fig. 6).

- *No quantum candidate.* Of the 619 points that become MPS-hard, 356 have $\eta \ge 0.3$; for none of them do both proxies stay below −10 until t_S. The largest proxy window among them is 1.0 (median 0.25): in non-classical quenches the proxies fire no later than the entanglement saturates, usually much earlier.
- *Near-classical windows are common.* Of the 201 MPS-hard points with $\eta \le 0.1$, 53% have a proxy window ≥ 2. The strongest is R1 (up to 38 at $\Omega/V_{nn} = 0.01$).

Per-family results of the screen are in App. D.

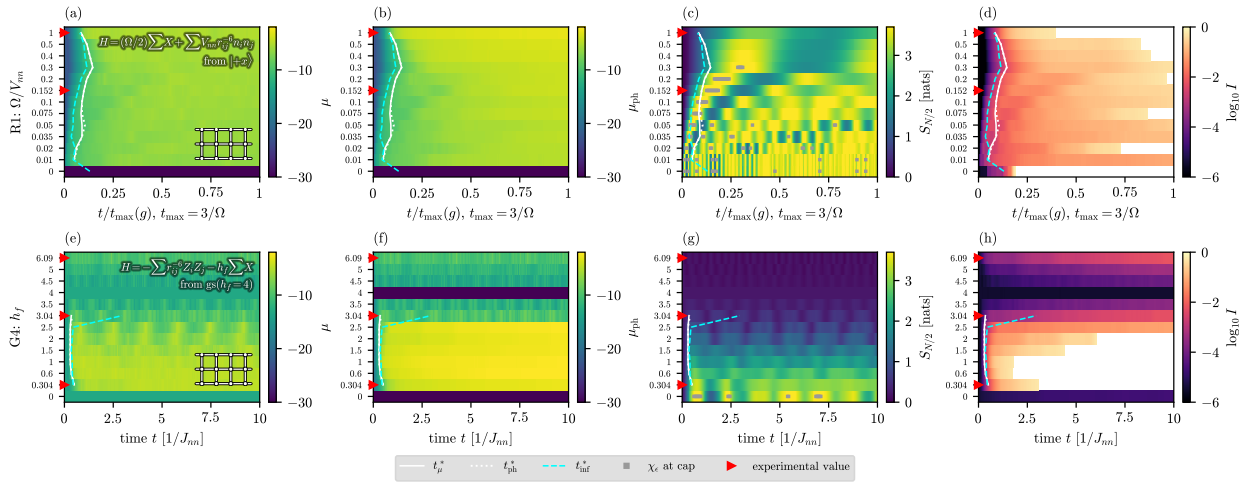


Figure 7: R1 (top) and G4 (bottom) over control parameter g (vertical, one strip per grid point) and time. Insets as in Fig. 1. Columns: μ, $\mu_{\rm ph}$, half-cut entropy S (grey squares: χ_ε at its cap), and $\log_{10} I$ for the $\alpha = 1$ RBM (white: run stopped at $I \geq 0.5$). Lines: t^*_μ (white solid), $t^*_{\rm ph}$ (white dotted), $t^*_{\rm inf}$ (cyan dashed). Red triangles: experimental or benchmark values of g. R1 is plotted against $t/t_{\max}(g)$, $t_{\max} = \min(300, \max(10, 3/\Omega))$.

9 R1: a near-classical RBM advantage

In the weakly driven 2D Rydberg Ising quench the RBM follows a volume-law state for up to $12\,t_S$ (Figs. 7, 8a–c).

- *RBM window.* At $\Omega/V_{nn} = 0.01$ the infidelity stays below 10^{-3} up to $t = 12$ ($\alpha = 1$, Jastrow), 15 ($\alpha = 2$) and 18 ($\alpha = 4$), while $S_{\rm rel} \geq 0.5$ from $t_S = 1.5$ on and reaches 1.13: an RBM window of 8–12. At $\Omega/V_{nn} = 0.05$ the RBM is accurate up to $t = 2.4$ ($\alpha = 1$), 3.0 ($\alpha = 2$) and 9.3 ($\alpha = 4$), an RBM window of 1.6–6. For $\alpha = 1$ the window closes at $\Omega/V_{nn} \approx 0.15$.

- *More hidden units help, a Jastrow factor does not.* Going from $\alpha = 1$ to 4 delays $t^*_{\rm inf}$ by 1.4–3.6× and lowers the early infidelity by 1–3 decades. The Jastrow factor alone is no better than $\alpha = 1$ ($\Omega/V_{nn} = 0.01$: $t^*_{\rm inf} = 10.5$ vs 13.5), and RBM×Jastrow is no better than the RBM alone (0.05: 2.4 vs 2.7). The error therefore comes from the weak drive, not from the Ising phase.

- *Larger cluster.* At $N = 16$, $t^*_{\rm inf}$ is equal or 10–20% shorter (0.01: 13.5 → 12; 0.05: 2.7 → 2.4; 0.3: 1.15 → 0.95), while t_S is unchanged and the proxy onsets move later (t^*_μ at 0.05: 5.4 → 9.6). The entanglement per boundary bond is the same for both sizes (App. E, Fig. 14), so the window survives the step in N but does not grow.

- *Proxies rank well but fire late.* $\mu_{\rm ph}$ predicts the infidelity (Spearman 0.92, also 0.92 at fixed S; μ: 0.85; S alone: 0.14), and $t^*_{\rm prox}$ orders $t^*_{\rm inf}$ perfectly across Ω ($\rho = 1.00$). But the RBM fails before the proxies at all 12 points (median $t^*_{\rm inf}/t^*_{\rm prox} = 0.77$), so the proxy window overstates the RBM window by ≈ 25%.

- *The window shrinks with* Ω. $t^*_\mu\Omega \approx 0.15$–$0.75$ for $\Omega/V_{nn} \leq 0.2$, while $t_S \approx 1.0$–1.5 is set by the Ising couplings; the proxy window closes at $\Omega/V_{nn} \approx 0.1$–$0.2$.

The same near-classical behaviour appears in Rydberg-dressed interferometry (R6, App. E), at the experimental soft-core radius with an RBM window of 1.5–4.5.

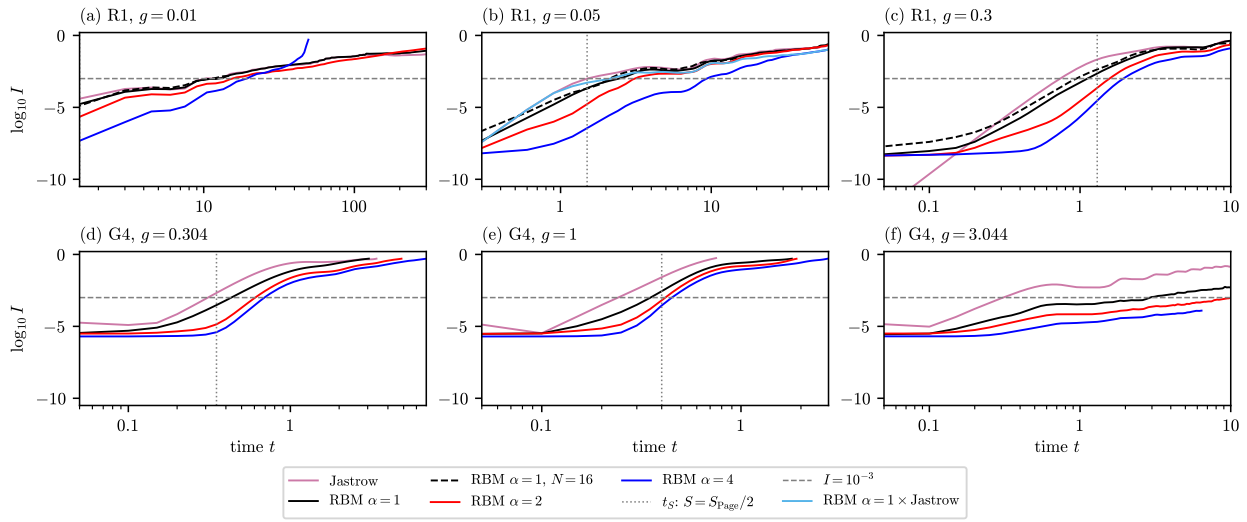


Figure 8: $\log_{10} I(t)$ per ansatz at three points of R1 (top) and G4 (bottom), logarithmic time axis. Dashed black: $\alpha = 1$ RBM on the 4×4 torus ($N = 16$). Dotted vertical: t_S (the state becomes MPS-hard at $N = 12$); dashed horizontal: $I = 10^{-3}$. Time in units of $1/V_{nn}$ (R1) and $1/J_{nn}$ (G4).

10 G4: no quantum RBM advantage

In the 2D TFI benchmark, proxy onset, RBM failure and entanglement saturation coincide; there is no window (Figs. 7, 8d–f).

- *The RBM fails at t_S.* For $0 < h_f \leq 2.5$ the $\alpha = 1$ RBM crosses 10^{-3} at $t = 0.35$–0.45. Where the state becomes MPS-hard ($h_f \leq 1$, $t_S = 0.30$–0.40), this is at or one save step after t_S; for $h_f \geq 1.5$ the entropy stays below half the Page value. Close to the initial field ($h_f \geq 3$) the quench is easy ($t^*_{\text{inf}} = 2.95$ at h_c, never for $h_f = 3.5$–5).

- *More hidden units barely help.* $\alpha = 4$ delays t^*_{inf} by at most $1.6\times$ ($h_f = 0.304$: $0.45 \to 0.70$; $h_f = 1$: $0.40 \to 0.45$).

- *Closest approach.* $h_f = 0.6$ ($\eta = 0.42$) and $h_f = 1$ ($\eta = 0.61$) are the only screened points with $\eta \geq 0.3$ where $S_{\text{rel}} \geq 0.5$ and $I < 10^{-3}$ hold at the same time, each only at the single save time $t = t_S$ ($h_f = 0.6$: $t_S = 0.35$, with $\alpha = 1$; $h_f = 1$: $t_S = 0.40$, with $\alpha = 2$ and 4). The proxies have already fired at 0.30–0.35, and I crosses 10^{-3} one step later. At $N = 12$, $S_{\text{rel}} = 0.5$ means $S \approx 1.8$ nats, so this is not a meaningful MPS-hard window. At $h_f = h_c/10$ ($\eta = 0.23$, below the non-classical threshold) the RBM window is 1.9 ($\alpha = 4$).

- *The proxies work.* μ_{ph} predicts the infidelity best (Spearman 0.94, 0.93 at fixed S; μ: 0.91; S alone: 0.29), and fires first (median $t^*_{\text{inf}}/t^*_{\text{prox}} = 1.33$; the RBM fails first at only 1 of 8 points). The onset ordering across h_f is $\rho = 0.64$ (t^*_μ) and 0.72 (t^*_{ph}).

- *Mechanism.* The same holds in every non-classical family (D3, R4; App. E): the proxies fire no later than t_S and the RBM fails at about the same time ($t^*_{\text{inf}} \approx 1.3\, t^*_{\text{prox}}$). On 12–16 sites, fast entanglement growth from a non-classical H_f comes with amplitude dynamics that fill the cluster spectrum of $\log|\psi|$, and that is what the RBM cannot follow (interpretation, not tested separately). The Clifford kicked Ising circuit, a natural candidate, fails for a different reason (App. E).

11 Verdict of Part II

Three of the four success criteria are met; the quantum RBM advantage is not (Table 3).

Table 3: Success criteria of `PLAN_rydberg_quench.md`.

criterion	result	met
Near-classical RBM advantage: $S_{\mathrm{rel}} \geq 0.5$ with $I < 10^{-3}$ ($\alpha \leq 2$ or RBM×Jastrow)	R1 $\Omega/V_{nn} = 0.01$: $t_S = 1.5$, $I < 10^{-3}$ up to $t = 12$ ($\alpha = 1$), 15 ($\alpha = 2$); R6: up to 2.4–3.6 ($\alpha \leq 2$, $t_S = 1.2$–1.5); also at $N = 16$	yes
Quantum RBM advantage: the same with $\eta \geq 0.3$ and $\mu, \mu_{\mathrm{ph}} < -10$	none; closest: G4 $h_f = 1$, one save step at t_S with proxies already above -10	no
Relevance: headline from Tier 1	R1 [1, 2] and R6 [4, 5] are Tier 1	yes
Proxy validity: Spearman ≥ 0.8 and partial ≥ 0.5 in ≥ 2 new families	μ_{ph}: R1 0.92/0.92, G4 0.94/0.93, D3 0.93/0.87, I1 0.82/0.80; fails in R4	yes

- Near-classical quenches with a weak drive ($\Omega/V_{nn} \lesssim 0.05$) are the regime where an RBM follows a volume-law state for several times t_S; a larger hidden-unit density helps, the exact Jastrow factor does not.
- The proxy window is the right screen for this regime (it selected R1, R6, I1), but it overstates the RBM window: in R1 the RBM fails at $\approx 0.8\, t^*_{\mathrm{prox}}$.
- In non-classical quenches the proxies, the RBM failure and the entanglement saturation coincide within a factor ≈ 1.3 on 12–16 sites; we found no counterexample.

Limitations.

- *Cluster size.* "MPS-hard" means $S_{\mathrm{rel}} \geq 0.5$ at $N = 12$–16, i.e. $S \gtrsim 1.8$–2.5 nats and $\chi_\varepsilon \leq 64$–256. The step $N = 12 \to 16$ leaves the entropy per boundary bond unchanged (Fig. 14); extrapolating it to a 10-site cut gives $S \gtrsim 6.8$ nats, $\chi \gtrsim 900$, which is a rough lower bound (not sure). The $N = 16$ RBMs ran only to $t = 15$ and at two families.
- *Phase proxy.* The time-unwrapped phase is not reduced modulo 2π, so μ_{ph} flags stabilizer states (S3) as dense. Near nodes (minlog ≈ -70, C2/R5) it changes by up to 6 under $dt/2$; t^*_{ph} moved in one of 48 checks.
- *TDVP.* The solver settings (App. F) were not varied. The $\Omega = 0$ runs show that TDVP error, not expressivity, can dominate (App. E). Zero initial hidden biases freeze the RBM for odd three-body terms (I5). Four $\alpha = 4$ runs stopped at the 6 h limit.

A Calibration: 1D TFI chain

The final code reproduces the calibration of the original 1D study (Fig. 9): the jump in μ is driven by amplitude zeros, not by entanglement. For $h_f = 0.5$:

- μ stays at -16 to -17 up to $t \approx 0.45$ and then jumps to ≈ -6 ($t^*_{\mu,\mathrm{rel}} = 0.54$);
- the jump coincides with a near-node of ψ: minlog reaches -22.9 at $t = 0.69$;
- the entanglement stays low throughout ($S < 0.8$).

The 1D picture is not purely "onset if and only if the quench crosses $h_c = 1$", however:

- $h_f = 1.2$ does not cross h_c, yet μ drifts up to -9;
- $h_f = 0.2$ does cross h_c, yet it stays below threshold until $t = 1.5$.

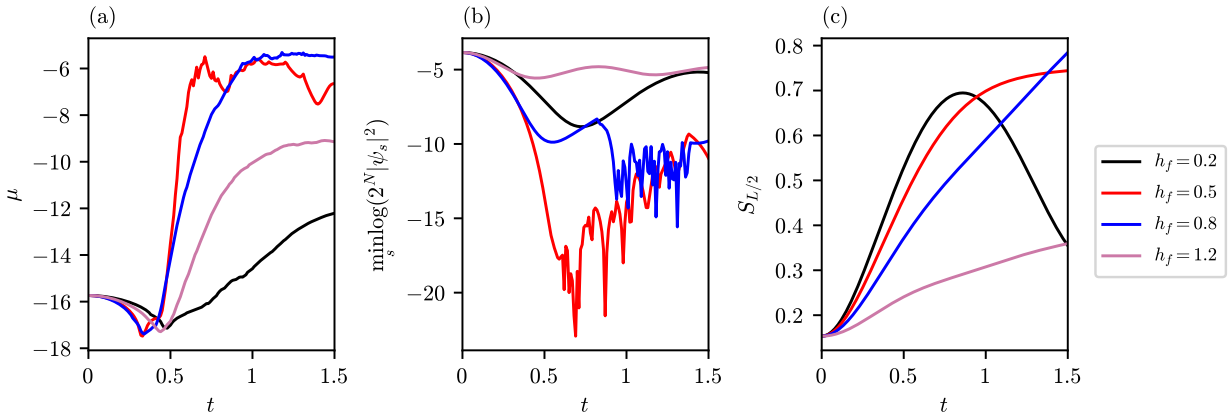


Figure 9: 1D TFI chain, $L = 12$ (open boundaries), quench from the ground state at $h = 1.5$ to h_f, analysed with the final code. (a) $\mu(t)$, (b) near-node indicator, (c) half-chain entanglement.

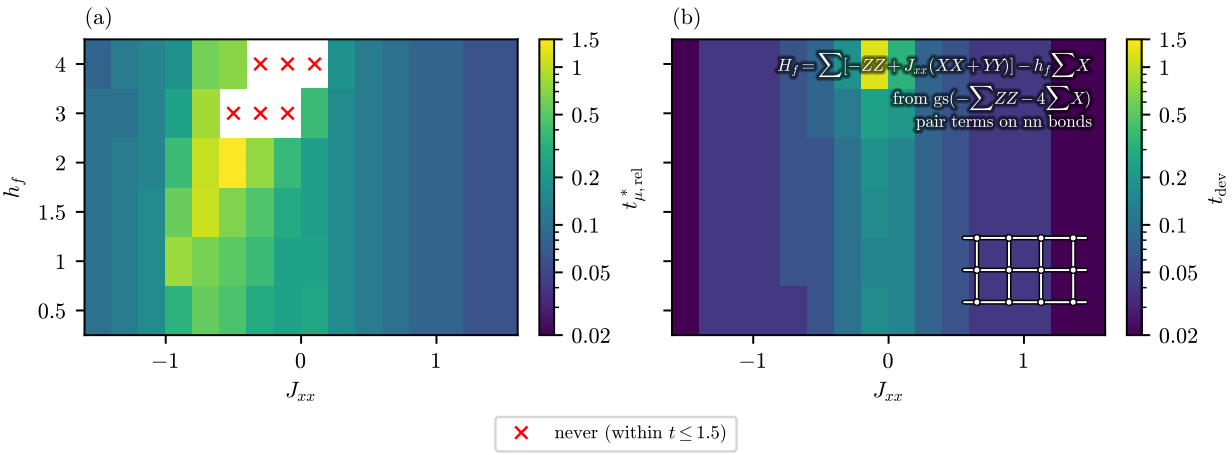


Figure 10: XY-coupling quench generalised to the (J_{xx}, h_f) plane (16×6 points, $dt = 0.02$). Insets as in Fig. 1. (a) onset time $t^*_{\mu,\mathrm{rel}}$. (b) plateau length t_{dev}. Red crosses: μ never crosses the threshold for $t \leq 1.5$.

B The (J_{xx}, h_f) plane

The plane scan (Fig. 10) confirms the expectation of the plan: a larger h_f widens the range of onset times but does not lengthen the plateau.

- larger h_f opens a "never hard" window at $J_{xx} \in [-0.5, 0.1]$ for $h_f \geq 3$;
- the longest finite onsets lie on the ferromagnetic side: $t^*_{\mu,\mathrm{rel}} \approx 1.0$–$1.5$ at $h_f = 1.5$–2 and $J_{xx} \approx -0.7$ to -0.5;
- the plateau length t_{dev} stays short ($\lesssim 0.15$) almost everywhere. The exception is the column $J_{xx} \approx -0.1$, which is close to the field quench, at large field: $t_{\mathrm{dev}} = 0.24$ at $h_f = 3$ and 1.2 at $h_f = 4$.

C Part II: setup details

Families. The 37 families (Table 4) were realised in experiments, serve as benchmarks, or are prominent in theory, sorted into three tiers: Tier 1 (realised in an experiment or an established numerical benchmark), Tier 2 (natural variations of Tier-1 set-ups or proposals) and Tier 3 (methodological controls). Each family has one control parameter g on a grid of 6–17 values that contains the experimental values (`rydberg/data/exp_params.csv`). Rydberg ground state: $|{\downarrow}\rangle \equiv |g\rangle$.

Hamiltonian parameters. Ω: Rabi frequency, Δ: detuning, V_{nn}: nearest-neighbour van der Waals interaction ($V(r) = V_{nn}(a/r)^6$), $R_b = (V_{nn}/\Omega)^{1/6}a$: blockade radius, R_c: soft-core radius of Rydberg dressing, κ in $1/r^\kappa$: interaction range of trapped-ion Ising models, J_z/J_{xy}: XXZ anisotropy, Θ: tilt of the quantisation axis of dipolar XY, θ: tilt angle of a product initial state from $|\uparrow\rangle$, h: longitudinal field, ε: small perturbation strength, J, b: Ising and kick angles of the kicked Ising circuit.

Clusters. $N = 12$ or 16: chains (open), rings, the 4×3 cylinder of Part I (`sq4x3`), the 4×4 torus (`sq4x4`), 4×3 and 4×4 triangular tori, a 2×2-cell kagome torus (`kag12`), a 12-site ruby strip (bond midpoints of four corner-sharing kagome triangles, open), a 12-qubit toric code on a 2×3 torus, and a 12-site honeycomb torus. Long-range couplings use minimum-image distances.

Initial states. Product states $|+x\rangle$, $|\theta\rangle$ (all spins tilted by θ from $|\uparrow\rangle$; $\theta = \pi/4$ unless θ is the control parameter), $|\mathbb{Z}_2, \theta\rangle$ (Néel tilted by θ), $|g\rangle$ tilted by 0.1–0.2 (Rydberg-native starts, tilted off the blockade zeros), and ground states of a pre-quench Hamiltonian. The tilts keep $\log|\psi|$ finite.

Exact evolution. Sparse Krylov evolution on 201 save times (21 for Floquet circuits, one per period), with the phase unwrapped along time (App. F). Halving dt at three points of every Tier-1 family (48 runs) changes ψ, S and μ by less than 10^{-9}. $\mu_{\rm ph}$ changes by up to 6 in two driven chains (C2, R5) where $|\psi|$ comes within e^{-70} of a node, but $t^*_{\rm ph}$ moves in only one of 48 checks, by one save step (C2: $6.0 \to 7.5$).

Triage. For every (g, t) we record the quantities of Sec. 2. A point is *MPS-hard (at this size)* from t_S on. The plan's classes, evaluated up to $t_{\max}$, are: (i) onset (μ crosses -10 after a low plateau), (ii) near-classical ($\eta \le 0.1$, $S_{\rm rel} \ge 0.5$, $\mu_{\rm ph} < -10$ throughout), (iii) quantum ($\eta \ge 0.3$, $S_{\rm rel} \ge 0.5$, $\mu, \mu_{\rm ph} < -10$ throughout), (iv) boring. Because every non-classical run eventually saturates the proxies on 12–16 sites, we also use a *window* version: a point is near-classical (quantum) if its proxy window is at least two, $t^*_{\rm prox}/t_S \ge 2$.

Deviations from the plan.

- Tier changes: I3 (no longitudinal field in the ion experiment [6]), D6 (the experiment is a 3D gas [7]) and R7 (no extractable parameters) moved to Tier 2; S8 (optional) was skipped. A new Tier-1 axis S3b follows the IBM circuit [8] ($J = \pi/4$, $h = 0$, kick $b = \theta_h/2$) on a 12-site ring (one heavy hexagon), a chain and the cylinder.
- R5 uses the experimental $V_{nn}/\Omega = 12$ [9] instead of 30; C2's control parameter is ω_m/Ω [10]; D4's is the staggered field δ with a balanced ± 1 pattern [11]; G3 starts from the ground state of the XY ferromagnet with a small transverse field, which removes the $U(1)$ zeros; I3 starts at $\theta = 0.1$.
- Tier-2 and Tier-3 families run on 12-site clusters only; disorder (R7, R8, D6) uses one realisation.
- Stage-2/3 RBMs run at $N = 12$; for families screened only at $N = 16$, the 4×3 cylinder was added to the screen.
- Grids have 13–17 instead of 15–21 points; the window classification is an addition to the plan's strict classes; the RBM budget (159 runs) exceeds the plan's 30 Stage-2 probes because Stage 3 used the full grids.

D Part II: screen of all families

Other Stage-3 families.

- *R0/R6, Rydberg-dressed Ising interferometry* [4, 5, 12]: $|+x\rangle$ evolves under the soft-core interaction $U_0/(1+(r/R_c)^6)$ (experiment: $R_c/a = 2.02$). Without a drive (R0) the phase is exactly two-body, so $\mu = \mu_{\rm ph} = -\infty$ and a Jastrow factor is exact, while S can reach the Page value. R6 adds a weak drive $\Omega = 0.05$.
- *R4, Rydberg arrays at fixed blockade radius* [13, 14, 15, 16]: quench at $\Delta = 0$, R_b/a across the experimental values 1.15 (checkerboard), 1.53 (ruby) and 2.4.
- *D3, Floquet-engineered dipolar XXZ* [17, 18]: from $|+x\rangle$, J_z/J_{xy} runs from Ising-like via XY to Heisenberg ($J_z/J_{xy} = 1$, where $|+x\rangle$ is an eigenstate); strongly non-classical ($\eta \approx 0.8$).
- *I1, trapped-ion long-range Ising* [19, 20, 21, 22]: ion chains realise $\sum_{ij} J_0|i-j|^{-\kappa} X_i X_j + B\sum Z_i$ with $\kappa \approx 0.6$–1.2, and dynamical quantum phase transitions, the 1D mechanism behind the jump of μ [23], were observed there. Here: $\sum r^{-\kappa} ZZ$ with a weak transverse field (0.1) from $|+x\rangle$; near-classical, range set by κ.
- *S3/S3b, kicked Ising near Clifford points* [8, 24, 25, 26, 27]: at $J = b = \pi/4$ the circuit is dual-unitary with maximal entanglement growth, yet from $|+x\rangle$ at $h = 0$ it produces stabilizer states (flat $\log|\psi|$, phase quadratic mod 2π): the most natural quantum candidate.

Screen results per family (Fig. 6).

- The largest η among MPS-hard points with both proxies still below -10 at t_S is 0.27 (I5, $h_x = 0.43$), where they fire only 0.05 later ($t_S = 0.25$, $t^*_\mu = t^*_{\rm ph} = 0.30$).
- Near-classical proxy windows ≥ 2: R1 (up to 38 at $\Omega/V_{nn} = 0.01$; ≥ 2 for $\Omega/V_{nn} \lesssim 0.1$), R6 (5–11 for $R_c/a \leq 1$ on the 16-site tori, 2.6–3.6 on the cylinder) and I1 (2–4.7 for $\kappa \geq 3$ on the 4×4 torus). R0 is exactly classical (window ∞) and reaches $S_{\rm rel} \approx 1$ on 2D clusters but stays at $S_{\rm rel} \leq 0.18$ on the chain, where the r^{-6} phases are effectively nearest-neighbour.
- *The plan's strict classes* are almost empty: over a full run up to $t_{\max}$ every non-classical point eventually crosses -10; only the exactly classical points (R0; R1 and R8 at $\Omega = 0$; Q1 at $\beta = 0$; G4 at $h_f = 0$; I2 at $h_x = 0$) are near-classical in the strict sense.
- *Kicked Ising.* At the Clifford point ($h = 0$) S3 has $\mu = -\infty$ at every save time on the 16-site chain and at 13 of 20 on the 12-site clusters ($\mu \approx -5$ at the others), but $\mu_{\rm ph}$ lies between -7.3 and -2.4 from the first save on: the stabilizer phase is a quadratic form only modulo 2π, and the time-unwrapped phase is not reduced modulo 2π.
- $\tilde{\mu}_{\rm ph}$ coincides with $\mu_{\rm ph}$ to plotting accuracy everywhere: the classical winding $\int E_d\, dt$ only changes coefficients of order ≤ 2, which the fit does not see.

The directory `rydberg/figs/screen/` holds quick-look sheets of every combination (μ, $\mu_{\rm ph}$, S, χ_ε fraction over (g,t)).

E Part II: all RBM runs

Protocol. Six Tier-1 families were taken to RBM+TDVP (Table 5): the three with the widest near-classical windows (R6, R1, I1) and three non-classical benchmarks (R4, D3, G4). Each runs on its 12-site cluster over the full screening grid (13–17 values of g) with an $\alpha = 1$ RBM; at three points per family (easy / onset / hard) we added $\alpha = 2$, $\alpha = 4$ and a pure Jastrow factor, at three near-classical points RBM×Jastrow, and at R1 and R6 the 4×4 torus ($N = 16$, $\alpha = 1$, $t \leq 15$).

Table 4: Studied families. η: median non-classicality over the grid; $S_{\rm rel}$: largest half-cut entropy relative to Page; "window": largest proxy window $t^*_{\rm prox}/t_S$ over the points with $\eta > 0$ (exactly classical points have an infinite window); "w.$_{\eta\geq.3}$": the same restricted to $\eta \geq 0.3$ (a quantum RBM advantage requires ≥ 2). Relevance: why the quench matters, with references.

ID	tier	H_f, control g	clusters	η	$S_{\rm rel}$	window	w.$_{\eta\geq.3}$	relevance
R0	1	Rydberg vdW, $\Omega = \Delta = 0$; tilt θ	1D, sq, tri, kag	0	1.09	∞	–	Ramsey interferometry in Rydberg-dressed lattices [4, 5, 12]
R6	1	dressing $U_0/(1+(r/R_c)^6)$, $\Omega = 0.05$; R_c/a	sq, tri	0.01	1.13	11	–	same, soft-core regime [4, 5]
R1	1	Rydberg vdW, $\Delta = 0$; Ω/V_{nn}	sq, tri, kag	0.04	1.13	38	0.54	2D Rydberg Ising quenches [1, 2]
G4	1	$-\sum r^{-6} ZZ - h_f \sum X$ from gs($h = 4$); h_f	sq	0.89	1.08	1.4	1.0	NQS benchmark of 2D TFI quenches [3]
R4	1	Rydberg, $\Omega = 1$, $\Delta = 0$; R_b/a	sq, kag, ruby	0.02	0.77	0.72	0.72	square/kagome/ruby arrays, toric-code spin liquid [13, 28, 14, 15, 16]
G1	1	Rydberg $R_b/a = 1.15$ from gs($\Delta = -2$); Δ	sq, kag	0.28	0.94	0.96	0.96	same [13]
R5	1	Rydberg, $V_{nn} = 12\Omega$; Δ/Ω	1D, sq	0.05	0.59	0.08	–	many-body scars after $\mathbb{Z}_2$ quenches [9, 29]
C2	1	as R5, $\Delta(t) = \Omega(0.85 + 0.98\cos\omega_m t)$; ω_m/Ω	1D	0.08	0.47	0	–	Floquet scar stabilisation [10]
D1	1	dipolar XY, axis tilted by Θ	sq, tri	1.0	0.99	0.75	0.75	spin squeezing in dipolar Rydberg arrays [30, 31]
G3	1	dipolar XXZ from gs(XY ferro$-0.2\sum X$); J_z/J_{xy}	sq	0.93	0.32	0	0	continuous symmetry breaking [32]
D3	1	dipolar XXZ; J_z/J_{xy}	1D, sq	0.82	0.96	0.67	0.67	Floquet-engineered XXZ [17, 18]
D4	1	AFM dipolar XY $+\delta\sum \eta_i Z_i$; δ	tri, kag	0.61	0.97	0.31	0.31	proposed dipolar Dirac spin liquid [11, 33]
I1	1	$\sum r^{-\kappa} ZZ + 0.1\sum X$; κ	1D, sq	0.07	1.12	4.7	–	trapped-ion long-range Ising [19, 20, 34, 21, 22, 35]
S3	1	kicked Ising $J = b = \pi/4$, field h	1D, sq	0.63	1.14	1.0	1.0	dual-unitary point [26, 27]
S3b	1	kicked Ising $J = \pi/4$, $h = 0$; kick b	1D, ring, sq	0.42	1.14	0.5	0.5	IBM Eagle "utility" circuit and its classical simulations [8, 24, 25]
Q1	1	QAOA layers (MIS phase, mixer β)	sq, kag	0.15	1.06	2.0	1.0	QAOA on Rydberg arrays and its NQS simulation [36, 37]
R2	2	Rydberg vdW, tilted/$\mathbb{Z}_2$ starts	sq	0.08	0.98	0.74	–	variation of R1
R3	2	Rydberg, $\Omega = 0.1$; Δ	sq	0.08	1.13	5.4	–	facilitation [38, 39]
C1	2	facilitation $\Delta = V_{nn} + \delta$	1D	0.18	0.27	0	–	same
R7	2	Rydberg, position disorder σ/a	sq	0.04	1.12	2.8	–	disordered Rydberg gases [39]
R8	2	Rydberg + random $\Delta_i \in [-2, 2]$	sq	0.07	1.13	8.0	–	variation of R1
R9	2	dual-species vdW; interspecies ratio	sq	0.05	1.13	2.4	–	dual-species arrays [40]
I2	2	dipolar Ising $\sum r^{-3} ZZ + h_x \sum X$	sq	0.34	1.10	1.7	1.0	variation of I1
I3	2	$-\sum ZZ - 0.25\sum X - h_z \sum Z$	1D	0.23	0.02	0	–	confinement [41, 6]
I5	2	$\sum_\triangle ZZZ + h_x \sum X$	tri	0.16	0.90	26	1.0	multi-spin terms in Rydberg gauge-theory proposals [42]
D2	2	dipolar XY ferro $+h_x \sum X$	sq	1.0	0.86	0.36	0.36	variation of D1
D5	2	dipolar XY $+\lambda\sum r^{-6} ZZ$	sq	0.45	0.93	0.67	0.50	mixed couplings
D6	2	diluted dipolar XY; filling	sq	1.0	0.61	0.14	0.14	disordered dipolar gases [7]
G2	2	Rydberg $\Delta = 3$ from checker-board gs; Ω	sq	0.11	0.58	0	0	ordered start
S1	2	toric code $+\epsilon\sum Z$	tc	0.69	0.77	0.14	0.14	toric code on processors [43]; exact RBM [44]
S2	2	cluster model $+\epsilon\sum X$	sq	1.0	1.02	0.33	0.33	cluster states [45, 46]
S4	2	kicked Ising $J = \pi/4 + \epsilon$	1D	0.62	1.14	0.33	0.33	leaving Clifford: magic [47]
I4	3	NN Ising $+h_x \sum X$	sq	0.36	0.41	0	0	control
S5	3	XY chain (free fermions)	1D	1.0	0.86	0.44	0.44	control
S6	3	$\sum XX + \Omega\sum Z$ (rotated basis)	sq	0.98	0.44	0	0	control
S7	3	XXZ chain	1D	0.94	0.73	0.17	0.17	control
C3	3	quantum East	1D	0.83	0.80	0.11	0.11	control

Further runs: R0 on the kagome cluster at $\theta = 0.892$ and $\pi/2$ (representability at $\Omega = 0$), S3 at three fields, and the two I5 points closest to a quantum RBM advantage. In total 159 runs (96 RBM $\alpha = 1$, 21 $\alpha = 2$, 19 $\alpha = 4$, 20 Jastrow, 3 RBM×Jastrow). Four $\alpha = 4$ runs (R4 at all three points, G4 at $h_f = 3.044$) hit the 6 h wall-clock limit at $t = 2.2$–6.4 with I still below 0.5.

Table 5: RBM runs per family (12-site clusters unless noted). "Ansatz points": where $\alpha = 2, 4$ and Jastrow were added.

ID	cluster, start	grid ($\alpha = 1$)	ansatz points	extra
R6	`sq4x3`, $\|+x\rangle$	13 values of R_c/a, $t \leq 60$	0.5, 1.0, 2.02	$N = 16$; RBM×Jastrow at 0.5
R1	`sq4x3`, $\|+x\rangle$	13 values of Ω/V_{nn}, $t \leq 3/\Omega$	0.01, 0.05, 0.3	$N = 16$; RBM×Jastrow at 0.05
I1	`sq4x3`, $\|+x\rangle$	13 values of κ, $t \leq 30$	1, 3, 6	RBM×Jastrow at 6
R4	`kag12`, $\|\theta\rangle$	14 values of R_b/a, $t \leq 10$	1.15, 1.53, 2.4	–
D3	`sq4x3`, $\|+x\rangle$	17 values of J_z/J_{xy}, $t \leq 10$	0, 0.75, 2	–
G4	`sq4x3`, gs($h = 4$)	13 values of h_f, $t \leq 10$	0.304, 1.0, 3.044	–

Results for the other families.

- *R6* behaves like R1 ($t_S = 1.2$–1.5; $I < 10^{-3}$ up to $t = 1.5$–2.7 with $\alpha = 1$ over the whole grid, 3.0–3.6 with $\alpha = 2$, 4.5–9.0 with $\alpha = 4$). At the experimental $R_c/a = 2.02$ [4] the RBM window is 1.5 ($\alpha = 1$) to 4.5 ($\alpha = 4$). Jastrow alone: $t^*_{\text{inf}} = 1.5$ vs 2.7 ($R_c/a = 0.5$); RBM×Jastrow: 2.7 vs 2.7. $\alpha = 4$ is not uniformly better: at 2.02 it becomes unstable at $t \approx 5$ (Fig. 12c). At $N = 16$: t^*_{inf} 2.7 → 2.4 (0.5), 2.4 → 2.1 (1.0), 1.8 → 1.8 (2.02).

- *D3, R4 (non-classical).* The $\alpha = 1$ RBM crosses 10^{-3} at $t = 0.15$–1.5, except near special points (D3 around $J_z/J_{xy} = 1$, where $|+x\rangle$ is an eigenstate). At the hard points larger α shifts t^*_{inf} by at most $\approx 2\times$. R4 is hard for all ansätze from $t \approx 0.2$–1 even where $\eta \leq 0.01$ ($R_b/a \geq 2$); the cause was not established.

- *I1 is near-classical ($\eta \leq 0.08$) but fails at t_S.* The $\alpha = 1$ RBM crosses 10^{-3} at $t = 0.45$–0.75, just at or after $t_S = 0.45$; $\alpha = 4$ extends this to 0.9–2.1; RBM×Jastrow at $\kappa = 6$: 0.6 vs 0.6. The long-range Ising phases ($r^{-\kappa}$ couplings to all sites) build up within $t \lesssim 1$, much faster than in R1/R6.

- $\Omega = 0$ *(R0, Fig. 13).* Let $n_{\text{eff}} = \#\{i < j : |\sin 2c^{\text{ph}}_{ij}| > 0.1\}$, with c^{ph}_{ij} the two-body coefficients of the phase: the number of pair phases that are not multiples of $\pi/2$, i.e. the number of two-body gates an RBM has to encode. The exact Jastrow state stays at $I \sim 10^{-16}$ for $t \leq 100$. The RBMs, which can represent it in principle, drift: at $\theta = \pi/2$, $t^*_{\text{inf}} = 52$ ($\alpha = 1$), 68 ($\alpha = 2$) and 44 ($\alpha = 4$); at $\theta = 0.892$ the $\alpha = 1$ RBM stays at $I \approx 3 \times 10^{-4}$ up to $t = 100$. Since n_{eff}/N reaches 5.5 (all pairs) by $t \approx 13$, $\alpha \gtrsim n_{\text{eff}}/N$ is neither necessary (at $\theta = 0.892$, $\alpha = 1$ suffices) nor sufficient (at $\pi/2$, $\alpha = 4$ fails first): the limitation at $\Omega = 0$ is the TDVP integration, not the expressivity.

- *S3 (Clifford kicked Ising).* μ_{ph} is above -10 from the first save (see the screen results above), and the $\alpha = 1$ RBM reaches $I = 0.95$–0.99 after one save step. Stabilizer states have exact RBM representations [44], so this is a failure of TDVP on the kicked protocol, whose cause we did not investigate.

- *I5*, the screen's best non-classical window, gives no RBM information: the RBM does not move. Its infidelity equals $1 - |\langle\psi_0|\psi_{\text{ED}}(t)\rangle|^2$ to three digits (0.067 at $t = 0.05$) for both $\alpha = 1$ and 2. With zero hidden biases $\log\cosh$ is even in s, so at initialisation the TDVP tangent space contains no odd three-body term $Z_iZ_jZ_k$. Random hidden biases would be needed to test I5 (not done).

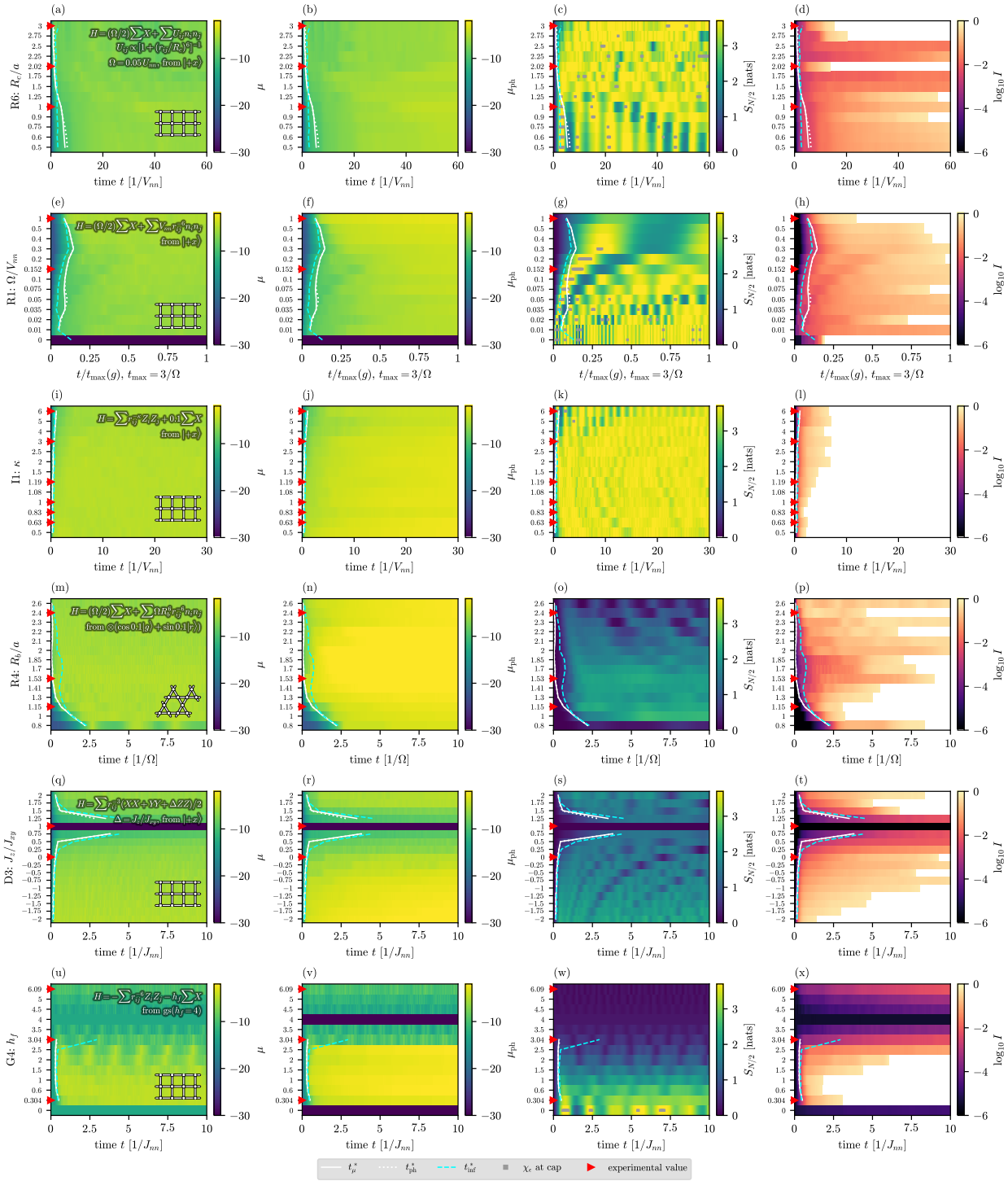


Figure 11: As Fig. 7, for all six Stage-3 families.

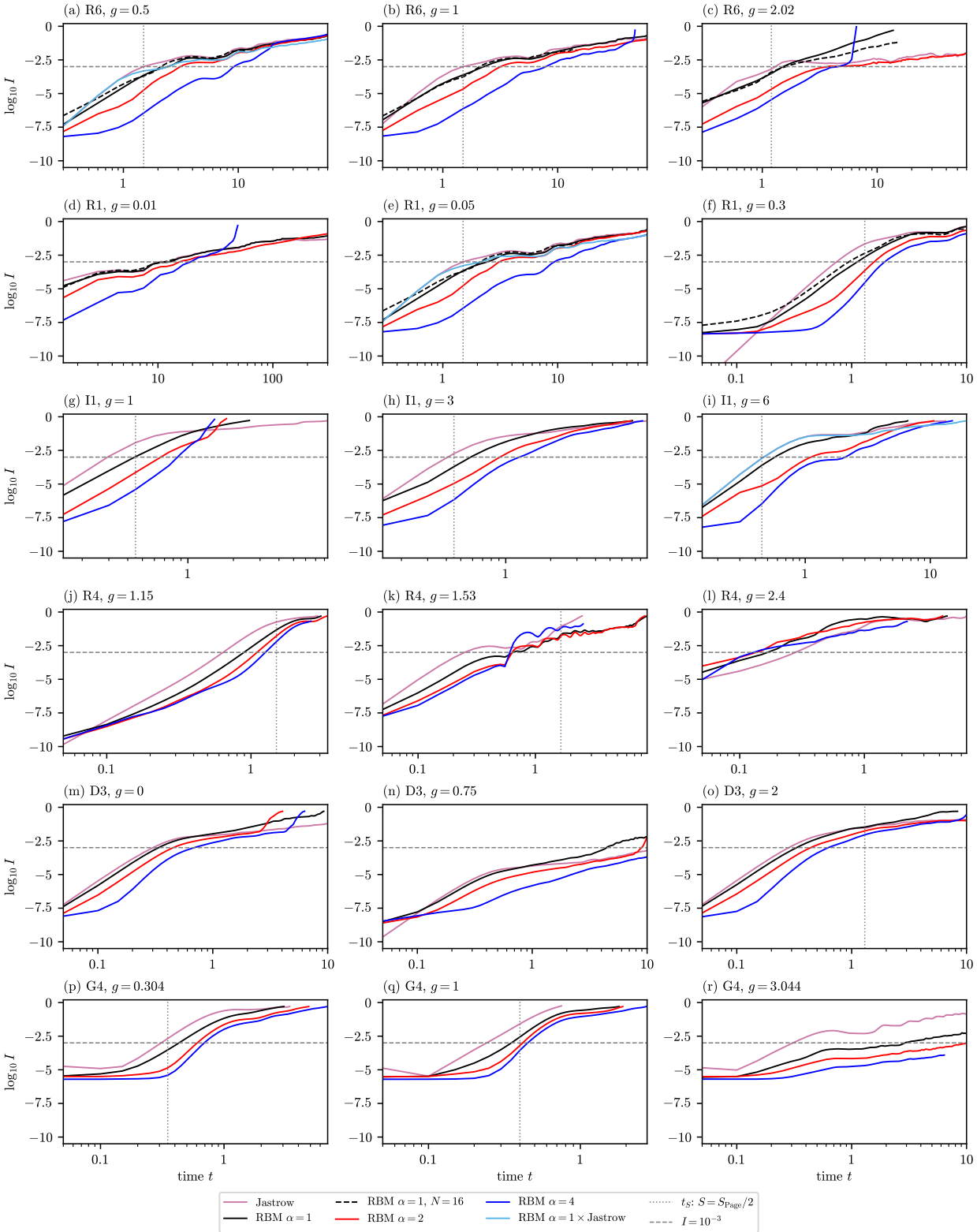


Figure 12: As Fig. 8, for all six Stage-3 families. Times in the units of Fig. 11.

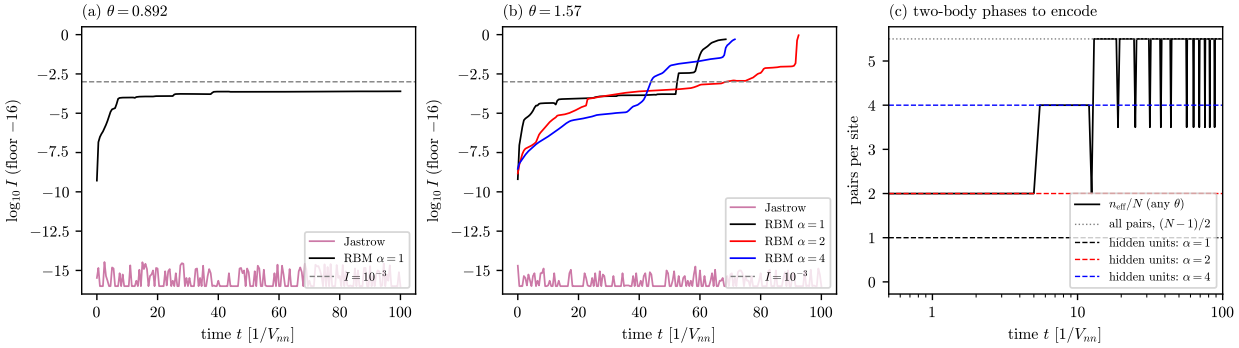


Figure 13: Representability at $\Omega = 0$ (R0, kagome, $|\theta\rangle$ start). (a,b) $\log_{10} I(t)$ per ansatz at $\theta = 0.892$ and $\pi/2$; the Jastrow factor is exact to machine precision. (c) Number of non-trivial pair phases per site, $n_{\rm eff}/N$, against the hidden-unit densities α.

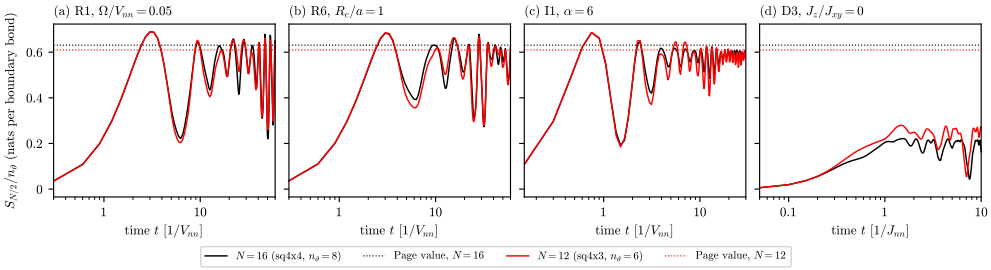


Figure 14: Half-cut entropy per cut boundary bond, S/n_∂, for $N = 12$ (`sq4x3`, black) and $N = 16$ (`sq4x4`, red) at four near-classical points; dotted: Page values per bond.

Proxy validity, all families (Tables 6, 7).

- $\mu_{\rm ph}$ is the best single predictor in D3, G4, R1 and I1 ($\rho = 0.82$–0.94; after controlling for S: 0.80–0.93). μ is close behind; S adds little once μ is known ($\rho(S|\mu) \leq 0.19$ in all six families), and neither does the χ_ε fraction.

- R6 is weaker (0.79/0.75, dropping to 0.62/0.39 after also controlling for t), and R4 fails ($\rho(\mu) = 0.10$): its infidelity rises within $t \lesssim 1$ for every R_b/a, while μ varies little.

- $t^*_{\rm prox}$ orders $t^*_{\rm inf}$ across g with $\rho = 0.97$ (D3), 1.00 (R1), 0.72–0.73 (R4), 0.64–0.72 (G4) and 0.56–0.59 (R6, I1). In the non-classical families (D3, G4, R4) the proxies fire first ($t^*_{\rm inf}/t^*_{\rm prox} \approx 1.3$–$3.5$); in the near-classical ones they fire late (R1 at all 12 points, R6 at 6 of 13).

- *MPS-hard / NQS-easy quadrant* (χ_ε at $\geq 90\%$ of its cap and $I < 10^{-3}$): 343 of 20356 points. Apart from the exactly classical points (R0, R1 at $\Omega = 0$, G4 at $h_f = 0$) they belong to R1 ($\Omega/V_{nn} \leq 0.3$, $t \leq 9$), R6 ($t \leq 9$), I1 ($t \leq 1.05$) and G4 at $h_f = 0.304$ ($t \leq 0.65$). The largest η in the quadrant is 0.23.

Table 6: Spearman correlation ρ of each proxy with $\log_{10} I$ ($\alpha = 1$, $N = 12$, all g and $t > 0$), and partial correlations controlling for S, for (S, t), and for μ. n: number of (g, t) points.

		ρ			$\rho(\cdot\|S)$		$\rho(\cdot\|S,t)$		$\rho(\cdot\|\mu)$	
family	n	μ	$\mu_{\rm ph}$	S	μ	$\mu_{\rm ph}$	μ	$\mu_{\rm ph}$	$\mu_{\rm ph}$	S
R1	2430	0.85	0.92	0.14	0.86	0.92	0.85	0.91	0.65	0.19
R6	2270	0.79	0.75	0.04	0.79	0.75	0.62	0.39	0.29	0.12
I1	396	0.78	0.82	0.37	0.75	0.80	0.71	0.72	0.43	0.19
R4	2000	0.10	0.38	0.17	−0.06	0.36	−0.07	−0.40	0.40	0.15
D3	2580	0.89	0.93	0.67	0.80	0.87	0.84	0.88	0.57	0.04
G4	1941	0.91	0.94	0.29	0.90	0.93	0.92	0.95	0.59	0.04
pooled	11965	0.82	0.83	0.32	0.80	0.84	0.83	0.85	0.39	0.19

Table 7: Onset ordering across g ($\alpha = 1$, $N = 12$). ρ: Spearman of t^*_μ (resp. $t^*_{\rm ph}$) with $t^*_{\rm inf}$ over the grid points where both are finite (n); last two columns: median of $t^*_{\rm inf}/t^*_{\rm prox}$, and the fraction of points at which the RBM fails before either proxy fires.

family	n	$\rho(t^*_\mu, t^*_{\rm inf})$	$\rho(t^*_{\rm ph}, t^*_{\rm inf})$	median $t^*_{\rm inf}/t^*_{\rm prox}$	RBM first
R1	12	1.00	1.00	0.77	100%
R6	13	0.56	0.56	1.33	46%
I1	13	0.59	0.59	1.50	23%
R4	14	0.72	0.73	3.50	0%
D3	16	0.97	0.97	1.29	0%
G4	8	0.64	0.72	1.33	13%

Table 8: Closest candidates to a quantum RBM advantage ($N = 12$). t_S: first time with $S_{\rm rel} \geq 0.5$ (∞: never); $t^*_{\rm inf}$ and $I(t_S)$ for the best ansatz tried.

ID	g	η	$S^{\max}_{\rm rel}$	t_S	t^*_μ / $t^*_{\rm ph}$	best $t^*_{\rm inf}$	$I(t_S)$	why it fails
S3	$h = 0$	0.61	1.14	2	2 / 2	2	0.99	$I \approx 1$ after one save step
S3	$h = \pi/4$	0.52	1.14	2	2 / 2	2	0.95	same
G4	$h_f = 1.0$	0.61	0.64	0.40	0.30 / 0.30	0.45	2.8×10^{-4}	proxies fire before t_S
D3	$J_z/J_{xy} = 2$	0.58	0.55	1.3	0.30 / 0.30	0.65	9×10^{-3}	RBM fails before t_S
D3	$J_z/J_{xy} = 0.75$	0.88	0.40	∞	3.75 / 3.85	> 10	–	never MPS-hard
I5	$h_x = 0.5$	0.31	0.48	∞	∞ / ∞	0.05	–	never MPS-hard; RBM frozen
G4	$h_f = 0.304$	0.23	0.95	0.35	0.50 / 0.50	0.70	3.9×10^{-6}	$\eta < 0.3$; proxy window 1.4
R1	$\Omega = 0.3$	0.15	1.06	1.3	1.45 / 1.45	1.95	3.0×10^{-5}	$\eta < 0.3$

F Numerical details

Part I, TDVP. Adaptive RK45 integrator with initial $dt = 0.01$, a dense holomorphic QGT (`QGTJacobianDense`) and NetKet's default linear solver, identical for every run.

Part II, exact evolution. Sparse Pauli-string Hamiltonians, evolved with `expm_multiply` from `scipy.sparse.linalg`, with sub-steps $dt \le \min(\Delta t_{\text{save}},\, 0.1/\max_s \sum_{s'} |H^{\text{off}}_{ss'}|)$. Piecewise-constant protocols (kicked Ising, QAOA, and the modulated drive of family C2, 24 pieces per drive period) are evolved segment by segment; diagonal segments exactly. The phase is unwrapped exactly for the diagonal part of each sub-step and by continuity for the rest: $\varphi \leftarrow \varphi + \arg(\psi_{\text{new}}\bar{\psi}_{\text{old}} e^{iE_d\,dt}) - E_d\,dt$. Lattice sites map to bit $N-1-i$ of the basis index, with $s_i = +1 \leftrightarrow |\uparrow\rangle$, as in NetKet.

Part II, TDVP. NetKet 3.19 TDVP with adaptive RK45 ($dt_0 = 0.01$; 10^{-4} for R4, which otherwise fails at $R_b/a = 2.4$), dense holomorphic QGT, the QGT error norm, and a smooth pseudo-inverse with $r_{\text{tol}} = 10^{-8}$. One TDVP driver per Floquet segment.

Software issues. NetKet 3.19's adaptive RK step recompiles a `lax.cond` at every step (unbounded memory); we patched it to run the branch eagerly, which leaves the step sequence unchanged. An early version of our save loop matched NetKet's `iter` stops (which use `np.isclose`, rtol 10^{-5}) to the exact save times and silently stopped recording; all affected runs were repeated with an explicit integration to each save time.

G Reproducibility

All code lives in `check_time_evolution/` and was run with the project `.venv` on CPU. Part I code is in `quench2d/` and Part II code in `rydberg/`. Each folder has its own `data/` and `figs/`. Run each script from its own folder; the paths below are relative to that folder.

- `quench2d.py`: lattice, dense and NetKet Hamiltonians, ED evolution, WHT and Fermi–Dirac analysis. `A0_FACTOR` switches A_0. Running `python quench2d.py` checks the FWHT, the SU(2) eigenstate, and that the NetKet and dense H agree.
- `ed_scan.py`: all ED scans and the 1D calibration, which write `data/ed_*.npz`. The scans are the XY-coupling quench (code label B), the field quench (A), the plane, $dt/2$, and $h_0 = 6$.
- `rbm_run.py FAM VAL ALPHA`: one RBM+TDVP run, which writes `data/rbm/rbm_*.npz`. The batch is `data/rbm/run_batch.sh` with the job list `data/rbm/jobs.txt`.
- `analysis.py`: writes `data/table_long.csv`, `table_tstar.csv`, `table_tstar_plane.csv` and `correlations.csv`.
- `make_figs.py`: writes every Part-I figure in `figs/`.
- `rydberg2d.py`: lattices, Rydberg/dipolar/Ising/circuit Hamiltonians, sparse evolution with phase unwrapping (`python rydberg2d.py` runs the self-checks).
- `rydberg_screen.py`: family specifications and the Stage-1 screen, writing `data/screen/<ID>_<lattice>_<init>.npz`; `rydberg_triage.py`: triage tables `data/triage*.csv`.
- `rydberg_rbm.py ID LAT INIT G ANSATZ`: one RBM/Jastrow+TDVP run, writing `data/rbm/*.npz`; batch runner `data/run_rbm_jobs.sh` with job lists `data/*jobs*.txt`.
- `rydberg_analysis.py`: `data/rbm_long.csv`, `rbm_tstar.csv`, `rbm_corr.csv`, `rbm_order.csv`; `rydberg_figs.py screen|map|scaling|main|omega0|ansatz`: the Part-II figures in `figs/` (`main` and `ansatz` also write the R1+G4 versions `*_hl`).

- References: `report/refs.bib`; experimental parameters: `data/exp_params.csv`.

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